ASL (American Sign Language) Recognition System

Project Overview

This project develops a real-time American Sign Language (ASL) recognition system that detects hand gestures and converts them into text**.** The system uses computer vision and a trained deep learning model to recognize ASL alphabet gestures captured through a webcam**.**

The model processes the webcam input, identifies the hand sign, and predicts the corresponding alphabet letter**.** This allows users to see the recognized gesture displayed as text in real time**.**

The main objective of this project is to demonstrate how Artificial Intelligence (AI) and Computer **Vision** can be used to interpret sign language, helping to improve communication between deaf or hard-of-hearing individuals and others.

Key Points:

- Real-time **ASL hand gesture recognition**
- Uses **Deep Learning and Computer Vision**
- Captures gestures using a **webcam**
- Converts **hand signs into text**
- Demonstrates how AI can assist **accessible communication**

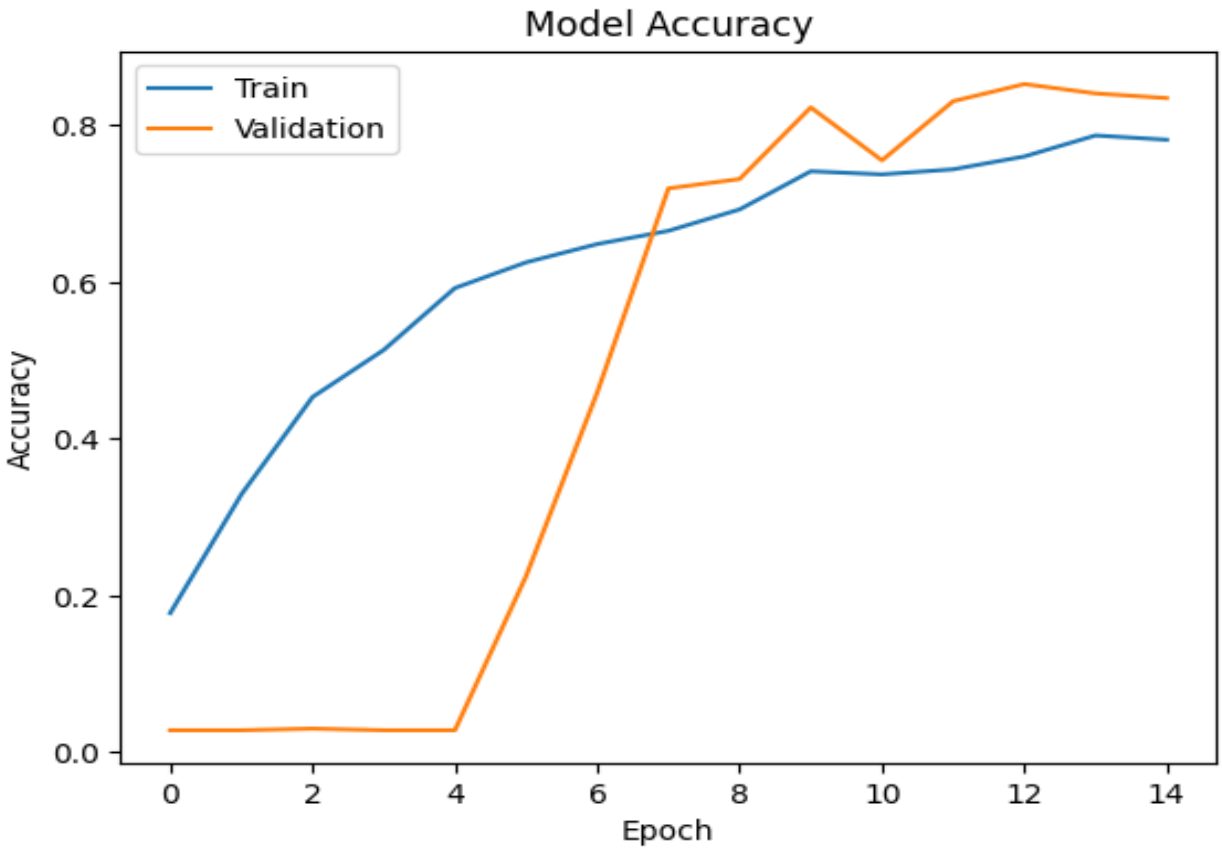
Dataset

The model is trained using the **ASL Alphabet Dataset**, which contains labeled images of American Sign Language hand gestures representing alphabet letters. These images help the model learn the visual patterns of different signs.

Dataset source:
https://www.kaggle.com/datasets/ayuraj/asl-dataset

Model Training

A Convolutional Neural Network (CNN) was used to train the model because CNNs are highly effective for image classification tasks. The model was developed using TensorFlow and Keras and trained for 15 epochs. After training, the model achieved good accuracy in recognizing several ASL alphabet gestures. The trained model was saved as 'asl_model.h5'.



Deployment Attempt

Initially, the model was deployed using Hugging Face Spaces to test the system online. Although the model was able to process images and generate predictions, real-time webcam recognition was not as smooth due to latency and browser limitations.

Local Implementation

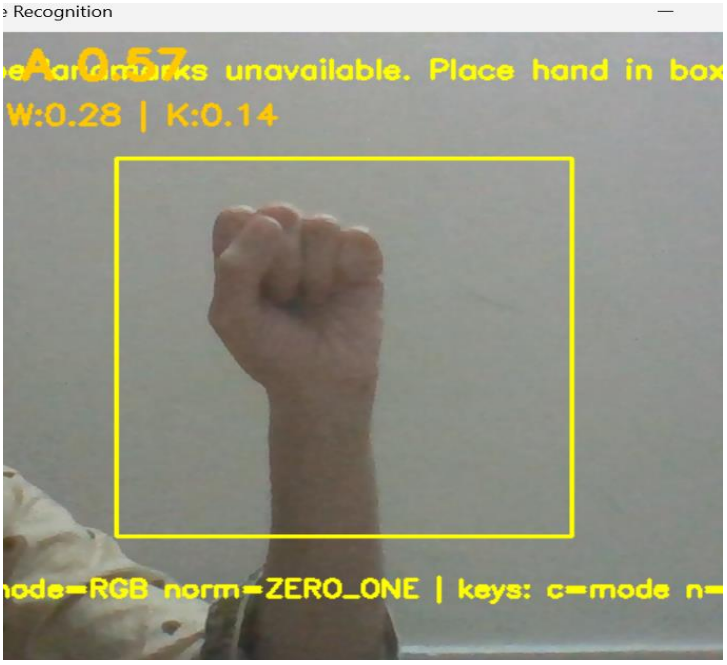
To improve real-time performance, the system was moved to a local implementation. Several libraries were installed including TensorFlow for model prediction, OpenCV for webcam access and image processing, MediaPipe for hand detection, and NumPy for numerical operations.

System Workflow

The system workflow is as follows: The webcam captures live video frames, MediaPipe detects the hand and extracts the hand region, the image is resized and preprocessed, and finally the trained CNN model predicts the corresponding ASL letter.

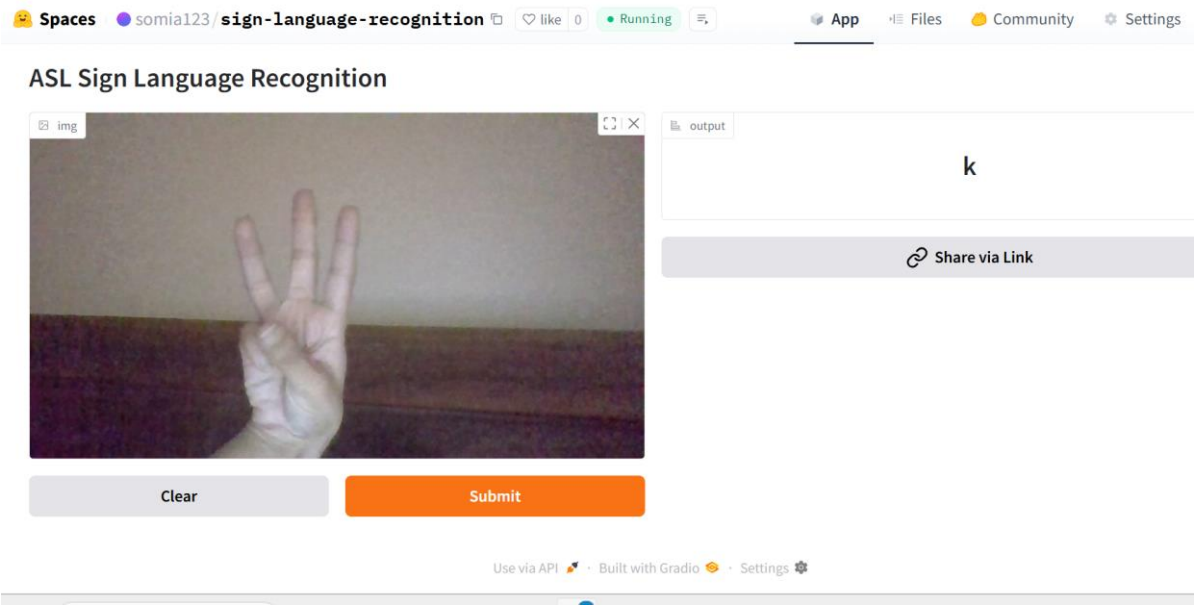
Results

During testing, the system successfully recognized several alphabet gestures. However, recognition accuracy was not perfect in all cases. Factors such as webcam quality, lighting conditions, and hand positioning affected the prediction results.



On Hugging Face

Not Correct Answer



Conclusion

This project demonstrates how computer vision and deep learning techniques can be used to build a real-time sign language recognition system. With improvements in dataset size, training time, and hardware quality, the system can achieve better accuracy and become more reliable for real-world applications.